

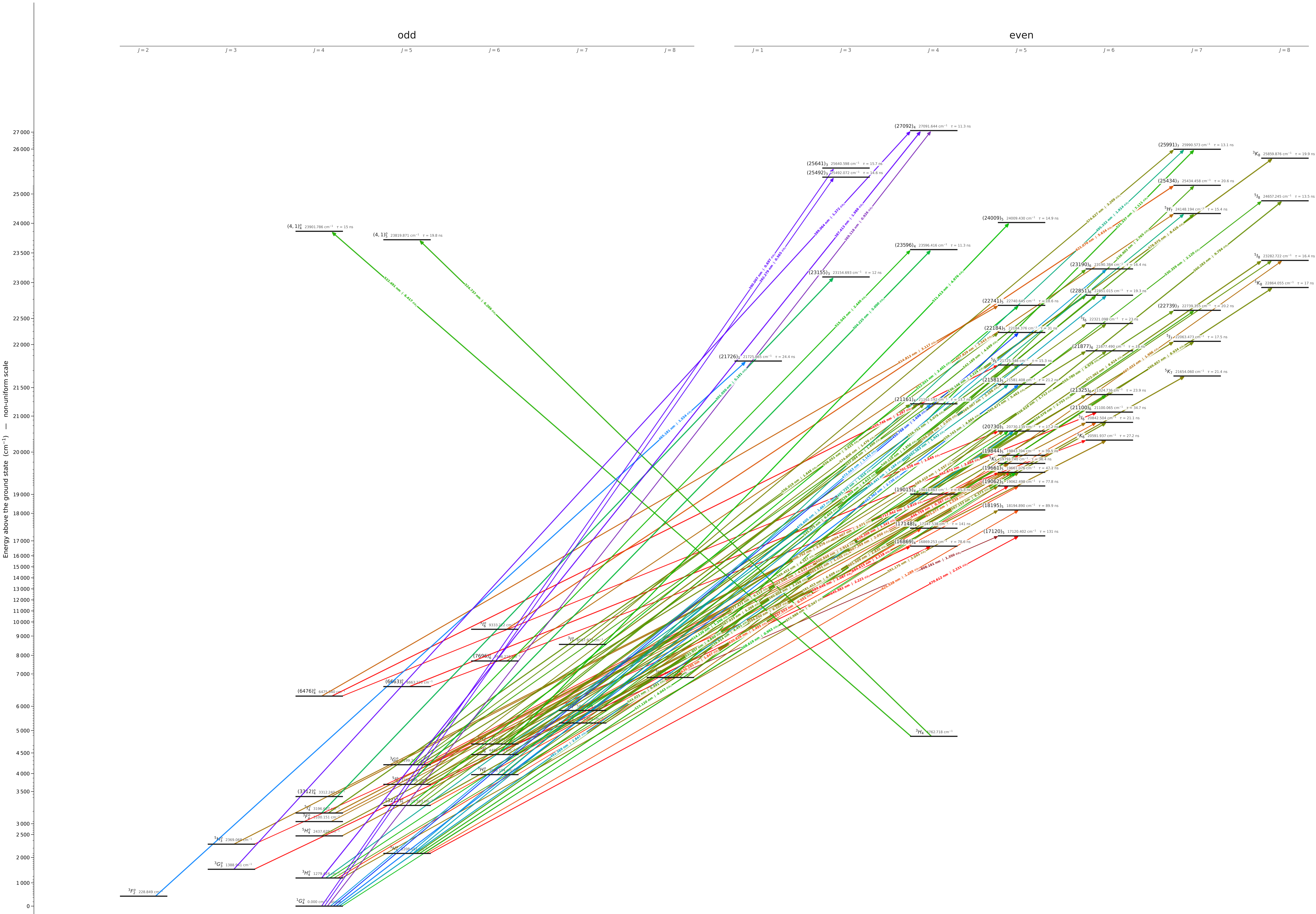
Cerium

Ce (Ce I). Strongest classified transitions below 2 μm (NIST): 67 levels, 90 lines.

Ce ⁺ ionisation limit 44672.00 cm⁻¹ = 5.53862 eV (223.854 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element |(U||er||U)| in ea0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨U er U⟩ (ea0)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
³ I _g - (23155) ₃	501.0501	500.9103	598.3284	5.191	6.200e+07	13.26	74.4	
³ I _g - (23596) ₄	504.2252	504.0846	594.5606	5.000	4.390e+07	14.08	49.6	
³ I _g - (24009) ₅	511.4128	511.2704	586.2044	4.976	3.410e+07	10.68	50.8	
³ I _g - (25991) ₇	521.3372	521.1920	575.0453	7.111	4.820e+07	12.15	63.1	
³ H ₄ - (4, 1) ₂ ^o	522.4915	522.3460	573.7748	6.417	6.500e+07	10.61	97.5	
³ H ₄ - (4, 1) ₃ ^o	524.7374	524.5913	571.3191	6.206	4.910e+07	8.038	87.2	
³ H ₇ - ³ H ₃	545.0754	544.9239	550.0018	6.403	3.420e+07	10.33	52.7	
³ I _g - (21161) ₁	556.6508	556.4962	538.5647	5.625	4.130e+07	11.62	56.6	
³ I _g - ⁵ I ₆	556.7513	556.5968	538.4674	5.076	2.750e+07	10.4	42.1	
³ I _g - ³ I ₆	560.2832	560.1277	535.0731	9.794	6.500e+07	11.79	87.8	
(3211) ₅ - ³ I ₆	567.1532	566.9958	528.5917	6.373	3.470e+07	7.543	73.2	
³ I _g - ⁵ K ₆	569.8573	569.6993	526.0834	8.934	5.140e+07	9.362	87.4	
³ H ₇ - ³ I ₆	572.0623	571.9036	524.0556	8.924	5.070e+07	9.705	83.2	
³ I _g - ³ K ₆	578.9750	578.8145	517.7986	8.426	4.360e+07	7.998	86.8	
³ I _g - ⁵ K ₇	581.4527	581.2915	515.5922	6.608	3.000e+07	7.437	64.2	
³ I _g - ⁵ K ₆	594.2502	594.0856	504.4886	6.107	2.770e+07	5.851	75.3	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).